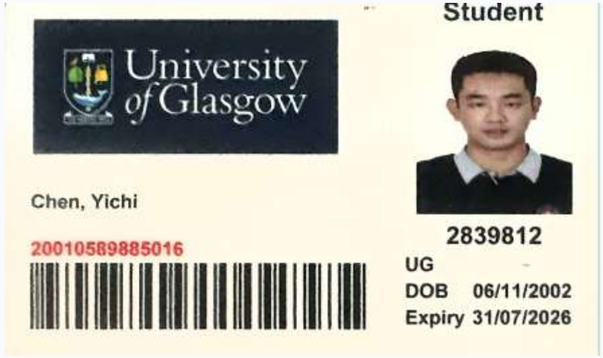


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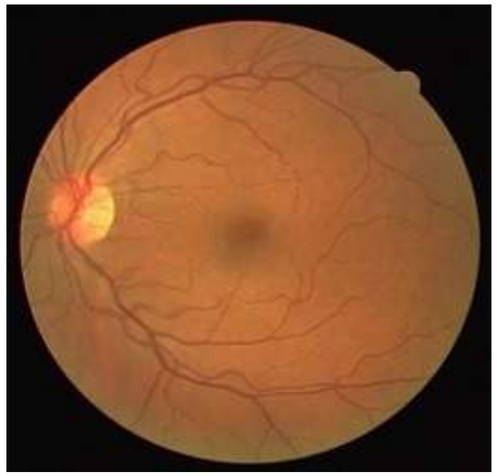
Final-Year Project Defense

Research on Fine Retinal Vessel Segmentation Based on nnU-Net

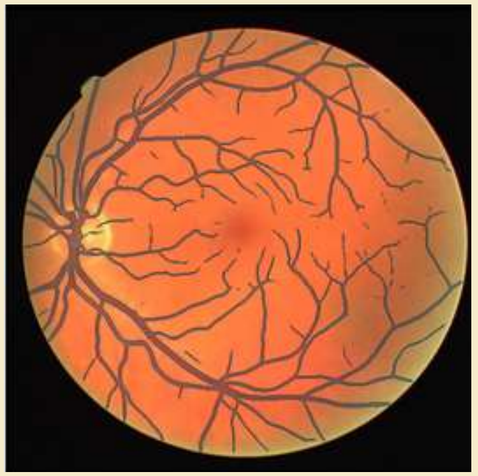
Chen Yichi | GUID: [redacted] | UESTC4006P

A controlled nnU-Net-based study focused on thin-vessel recovery, structural continuity, and false-positive control.

Original image → Vessel segmentation



Original image

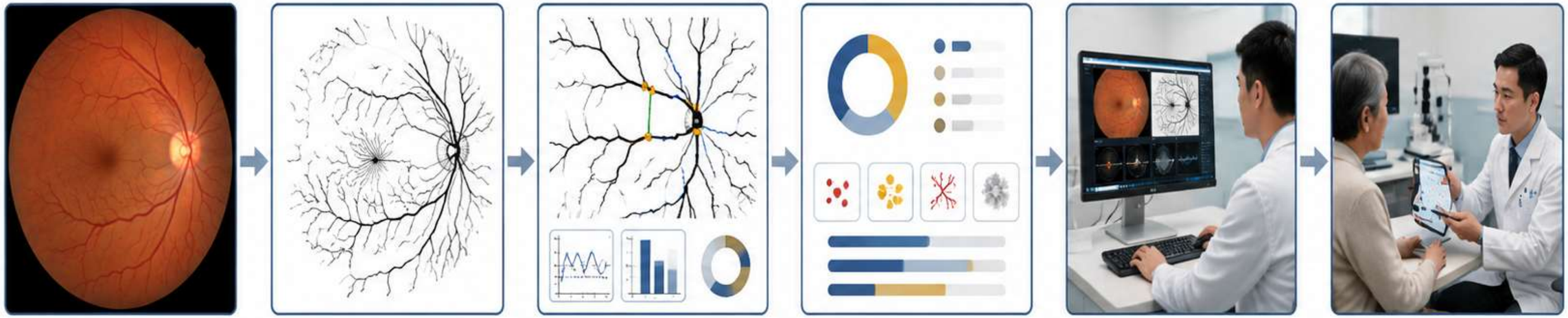


Vessel segmentation

The project studies how nnU-Net can better recover fine retinal vessel structure.

Clinical Significance and Use Scenario

Clinical support, not automatic diagnosis.

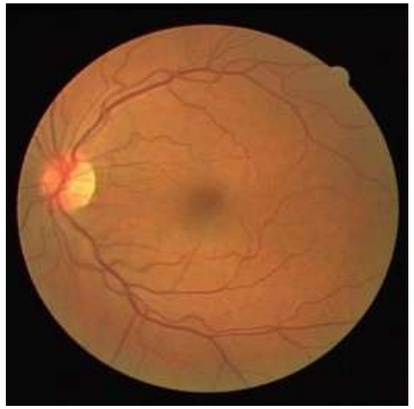


Fundus Image → **Vessel Segmentation** → **Vascular Feature Extraction** → **CAD Risk / Disease Suggestion** → **Doctor Review** → **Patient Management**

Segmentation provides the measurable vessel map used by later CAD modules.

Final diagnosis and patient decisions remain under doctor supervision.

Task Challenges in Fine Vessel Segmentation



Original fundus image

Fine peripheral vessels are weak and partly hidden by background texture.



Contrast-enhanced view

Preprocessing can make vessels clearer, but segmentation still needs structural modeling.

- Thin peripheral vessels are low-contrast and easily confused with background texture.
- Curved branches, bifurcations, and local discontinuities require structural continuity modeling.
- Higher sensitivity often introduces scattered false positives, so SE and PR must be balanced.
- A mature baseline such as nnU-Net leaves limited room for superficial improvements.

Core research question: what structural changes still produce useful gains on top of nnU-Net?

Experimental design

Baseline and Overall Technical Route

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Stage 1

Baseline setup

nnU-Net v2 establishes the controlled reference.

Preliminary exploration

CLAHE and Focal Loss affect convergence but do not improve final performance.

Stage 2

PA-Net migration

LPT and AFFM are transferred and screened under nnU-Net.

LPT mainline

Deformable sampling, multi-scale aggregation, CA, and DSConv are tested.

Stage 3

Output refinement

Connected-component filtering targets small isolated false positives.

Result analysis

Models are compared by SE, PR, SP, ACC, and F1-score.

512 x 512

Patch size

2D nnU-Net v2

5-fold

Validation

Unified later-stage protocol

100 epochs

Training

Most structural experiments

First structural screening

PA-Net Migration and Module Screening

Ablation snapshot

Unified 5-fold validation

Model	SE	PR	F1-score	F1 change
nnU-Net v2 Baseline	0.7627	0.8238	0.7891	Reference
PA-Net Full	0.7211	0.8570	0.7782	-0.0109
LPT Only	0.7597	0.8361	0.7935	+0.0044
AFFM Only	0.7573	0.8400	0.7942	+0.0051

0.7942
Best transferred F1
AFFM only

+0.0044
LPT branch gain
over baseline F1

-0.0109
Full migration
below baseline F1

What the screening shows

Direct transplantation was not enough.

- Full PA-Net increased PR, but sacrificed SE and reduced F1-score.
- Single-module variants were more stable under the same protocol.
- LPT remained the clearest route for later bottleneck structural modeling.

Design implication: keep the LPT-centered route, then test finer vessel-aware modules.

Screening criterion
F1 gain without clear SE loss

Next route
LPT-centered modeling

Module-level decision after screening

PA-Net Full
Not retained
Whole-branch migration lost sensitivity and F1-score.

LPT Only
Retained
Most relevant structural path for bottleneck redesign.

AFFM Only
Reference only
Slight F1 gain, but less aligned with later fine-vessel focus.

LPT, Deformable Sampling, and Multi-scale Modeling

The mainline gradually shifts from fixed high-level interaction to vessel-aware sampling.

Fixed LPT
Original high-level structural module.

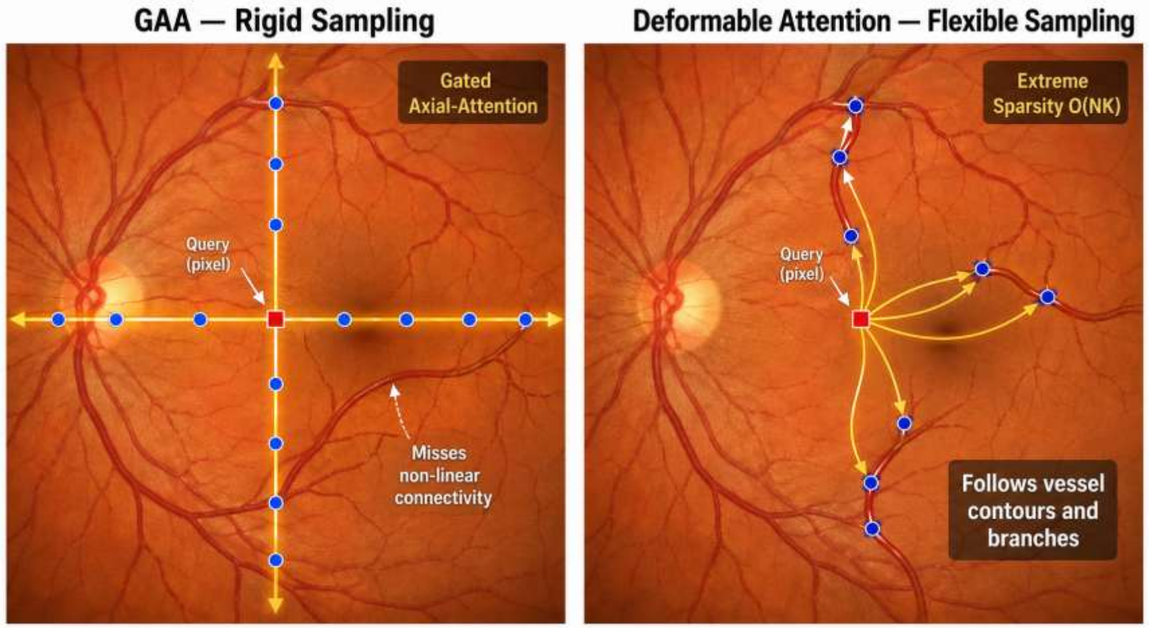
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Deformable
Offsets adapt sampling to curved vessels.

->

Multi-scale
Cross-level features support both detail and context.

- 4-point deformable LPT improves F1-score to 0.7958.
- Modulation increases detection tendency but can reduce precision.
- Multi-scale sampling improves overall balance by reusing shallow detail.



Coordinate Attention and Dynamic Snake Convolution

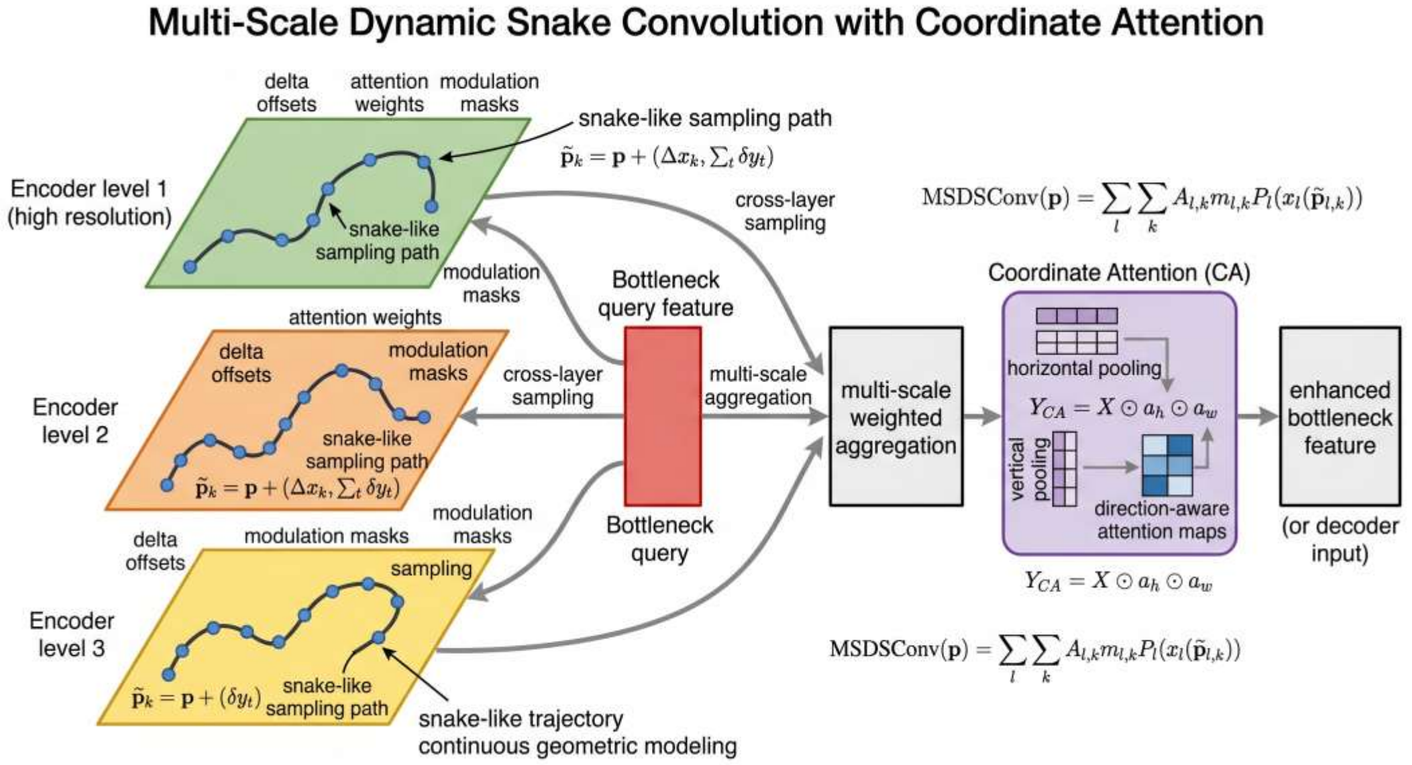
Coordinate Attention

Adds direction-aware positional recalibration after multi-scale aggregation.

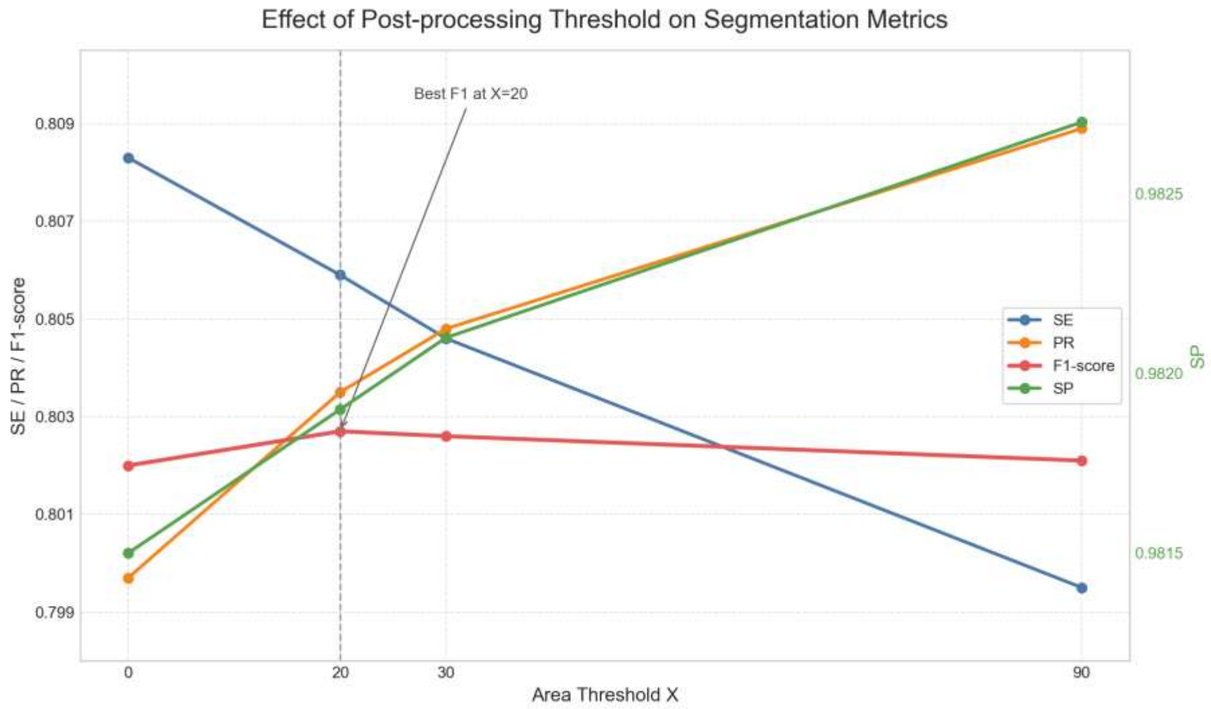
Dynamic Snake Convolution

Uses continuous trajectory-style sampling to better follow slender and curved vessels.

Observed effect: SE rises strongly, but PR drops, revealing a clear recovery vs false-positive trade-off.



Post-processing and clDice Exploration



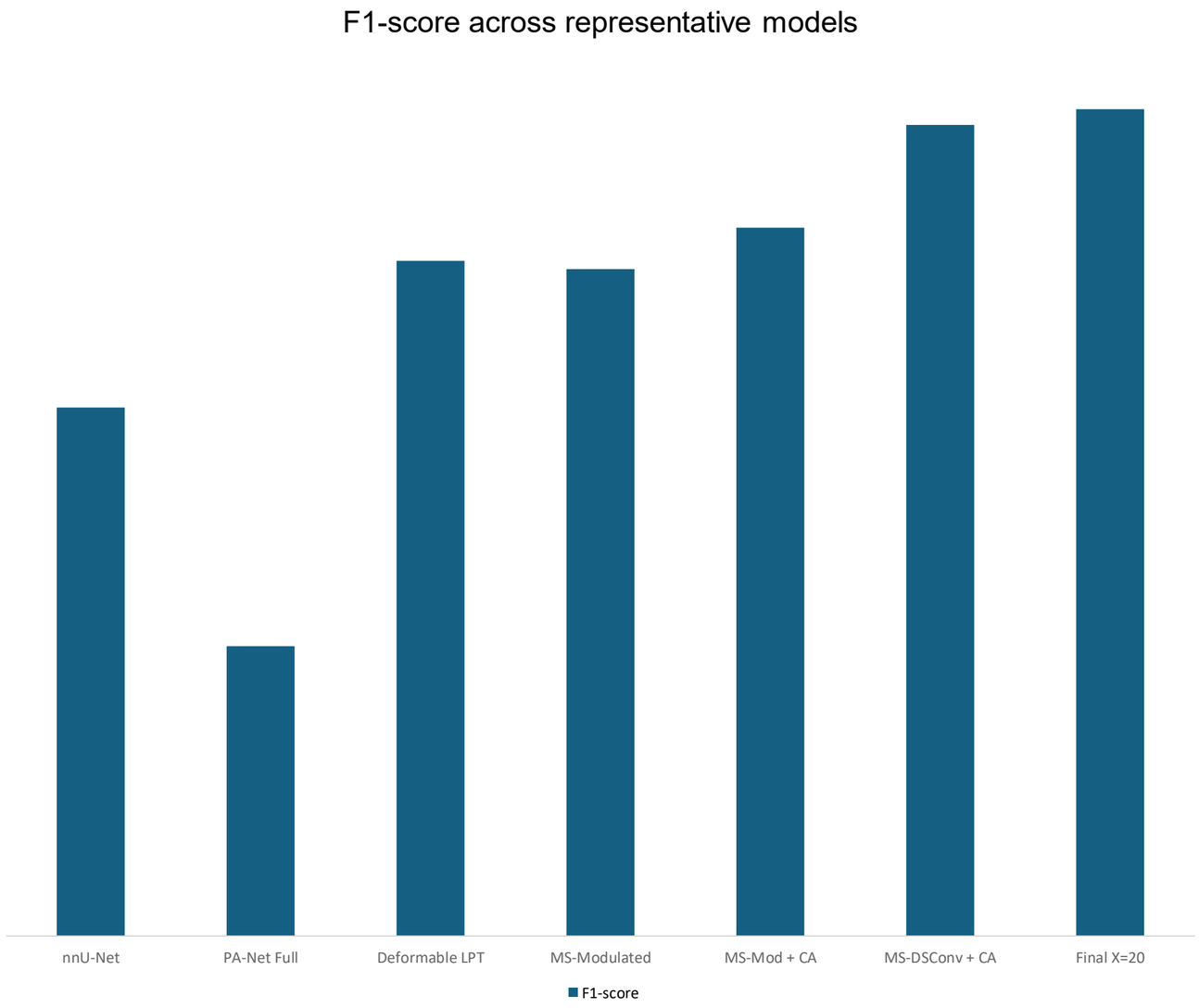
Two refinement ideas address what remains after MS-DSConv + CA.

Retained: connected-component filtering at X = 20 removes small isolated false positives and gives F1 = 0.8027.

Explored but not retained: clDice increases sensitivity in some settings, but lowers PR and F1-score under the representative X = 20 setting.

Unified later-stage comparison

Quantitative Results



0.8027

Final F1-score

MS-DSConv + CA, X = 20

+0.0136

Absolute F1 gain

vs. later-stage nnU-Net baseline

0.8059 / 0.8035

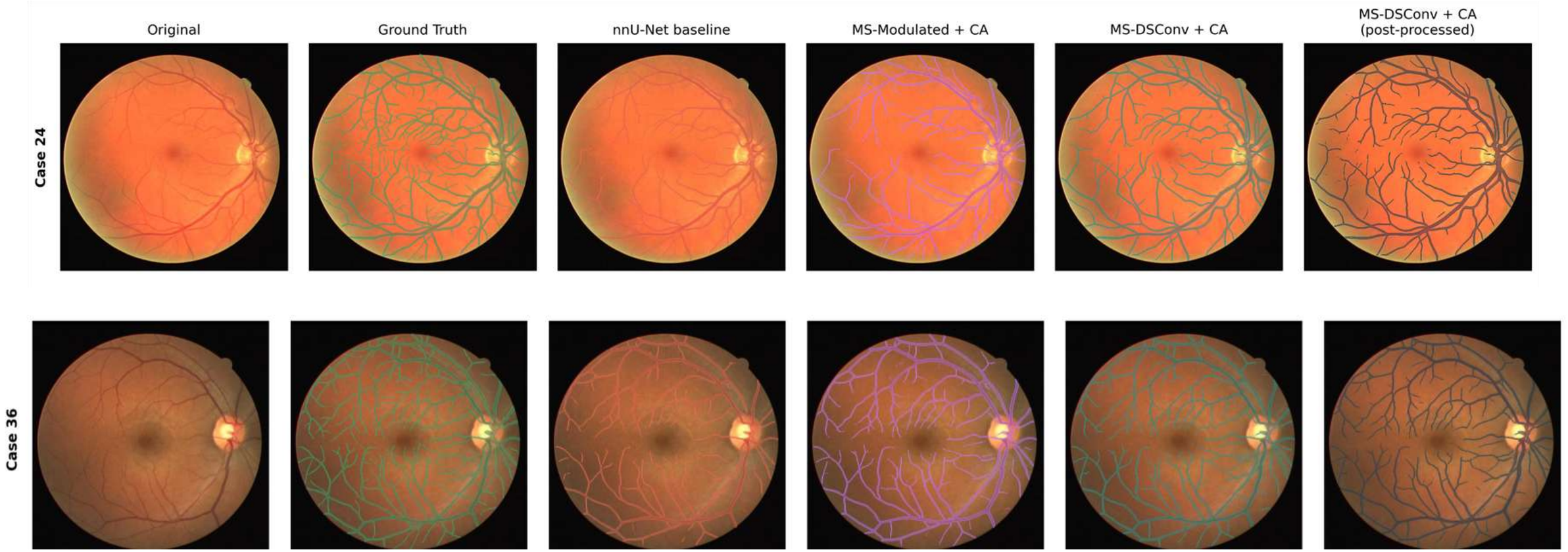
Sensitivity / Precision

balanced after light filtering

The improvement is modest, but it is obtained under the same later-stage protocol and follows a consistent SE-PR trade-off.

Representative prediction behavior

Qualitative Results



Baseline predictions are cleaner but often miss weak fine vessels.

MS-DSConv + CA recovers more continuous branches, with post-processing reducing isolated noise.

What remains open

Limitations and Future Work

Current limitations

- Validation is still limited to public benchmark conditions.
- False-positive control partly relies on rule-based post-processing.
- DSConv, CA, and multi-scale interactions need deeper interpretability analysis.
- Fine branches and low-response regions remain difficult.

Defense position: the method is a research-stage improvement, not a finished clinical product.

Future work

- Analyze nnU-Net preprocessing specifically for retinal images.
- Move false-positive control into the network or training objective.
- Test broader data across devices, pathologies, and annotation styles.
- Explore a more coordinated encoder-decoder structural design.

Research direction: move from benchmark gains toward robustness, explainability, and use-case evidence.

Next validation route

1 Broader data

Evaluate cross-device and cross-pathology robustness beyond the current benchmark setting.

2 Model-integrated control

Reduce isolated false positives through training objectives or learned refinement.

3 Clinical-style evidence

Assess reliability with expert review, failure cases, and diagnosis-oriented use scenarios.

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Conclusion

A modest but stable improvement path was identified within nnU-Net.

Final retained scheme: MS-DSConv + Coordinate Attention + light connected-component post-processing.

0.8027

Final F1-score

vs. 0.7891 baseline

Q&A

Thank you.